

- 3** Endurance athletes may suffer from muscle fatigue when contraction of the muscle fibres is disrupted.

It is thought that the disruption of the contraction of muscles is caused by a lack of calcium ions within the muscle fibre.

- (a) Explain why a decrease in calcium ions could disrupt the contraction of muscles.

(3)

- (b) The body controls the concentration of calcium ions circulating in the blood.

This concentration is kept within the range 80 to 100 mg dm⁻³.

- (i) Name the type of process involved in maintaining calcium ion concentrations within this range.

(1)

- (ii) The calcium ion concentration in the blood is kept within a range of 6% above and below the mean concentration.

The mean calcium ion concentration in the blood of a person was 91 mg dm^{-3} .

Calculate the **maximum** possible difference in the calcium ion concentration in the blood of this person.

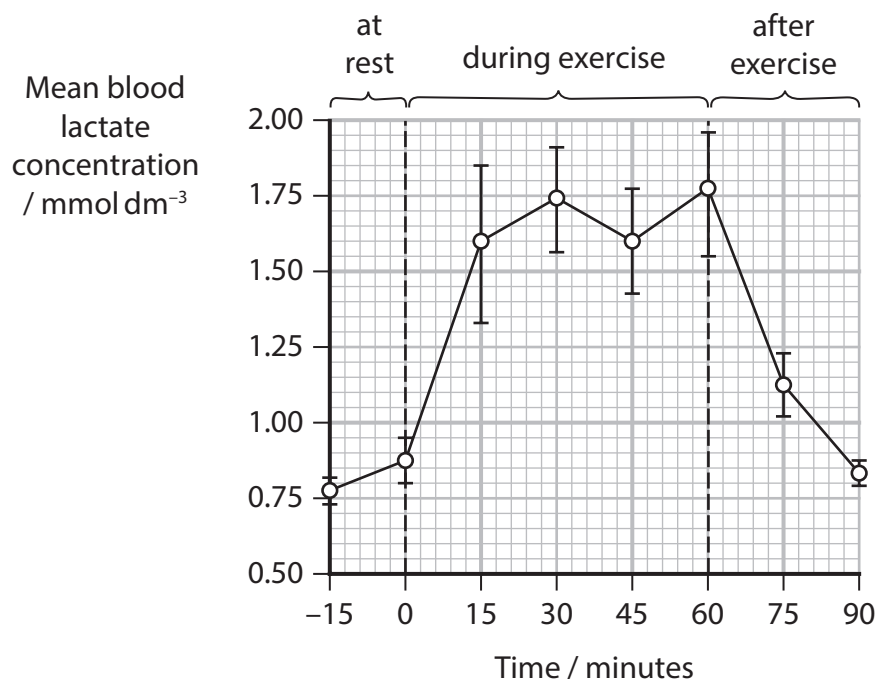
(2)

Answer mg dm^{-3}

- (c) In an investigation, a group of cyclists completed a 30 km time trial in a simulator.

The blood lactate concentration was measured at rest, during exercise and after exercise.

The results are shown on the graph.



- (i) Explain the effect on the lactate concentration from 15 minutes to 60 minutes on the graph.

(2)

- (ii) State **one** reason why the lactate concentration was measured 15 minutes before starting the exercise.

(1)

- (iii) State **one** variable that would need to be controlled in this investigation and how this would be achieved.

(1)

(Total for Question 3 = 10 marks)